



Brand equity in the business-to-business market

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Abstract

Brands have been developed by consumer companies but have been slow to develop in business-to-business marketing. This article explains the concept of brand equity in a specific industrial marketing setting. In addition, the sources of brand equity are investigated as well as the appropriate communications strategy and the relative importance of brand relative to other purchase criteria. The research method used was a conjoint analysis experiment. The subjects were decision-making unit (DMU) members of industrial companies in South Africa that purchase medium-voltage electrical equipment. Research results suggest that while brand equity has a role to play, price and delivery were more important. However, a price premium can be obtained when a company has high brand equity. Implications for managers are discussed. © 2003 Elsevier Inc. All rights reserved.

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1. Introduction

While brands and their management have dominated the marketing of goods and services to consumers, the idea has been slow to take hold in business-to-business marketing. This problem rests partly in the belief that because brands are irrational, they have little significance when dealing with a corporate entity that makes buyers' decisions on a rational basis (Rosenbroijer, 2001).

Most discussions of marketing in technical fields focus on the performance characteristics of the product or on the needs of buyers addressed by tangible features of the product (Shaw, Giglierano, & Kallis, 1989). However, studies do point out cases where price and the hard tangible factors of the physical product do not fully explain the purchase decision. Intangible aspects such as overall supplier reputation matter even in rational and systematic decision making (Mudambi, Doyle, & Wong, 1997).

The question that arises is whether the industrial buyers who are rational trained professionals and who normally operate within buying centers can be influenced by brand images that are based on nonfunctional and subjective attrib-

utes. According to Gordon, Calantone, and di Benedetto (1993), business-to-business product and service providers stand to gain sustainable competitive advantages through the development and strategic use of brand equity, particularly when competing in today's global economy.

The objectives of this study are to explore the existence of brand equity in a specific business-to-business product setting and to investigate the sources of brand equity and its appropriate communication channels and the relative importance of brand relative to other purchase decision criteria.

2. Theoretical background

2.1. Brand equity

Brand equity is derived from the overall brand image created by the totality of brand associations, perceived by customers (Michell, King, & Reast, 2001). Therefore, the attainment of a positive image on core values and any other values that differentiate it should be of the highest priority to any company (Hague & Jackson, 1994). Aaker (1996) identified four major sources of brand equity as brand loyalty, brand awareness, perceived quality, and brand associations, while Keller (1998) combines the sources of brand equity into brand awareness and brand image.

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The competitive advantage of firms that have brands with high equity include the following: a price premium can be attained; increased demand by customers; brands can be extended easily; communications will be more readily accepted; there will be better trade leverage; larger margins could be obtained; and the company will be less vulnerable to competitive marketing actions (Aaker, 1996; Hague & Jackson, 1994; Keller, 1998; Quelch & Harding, 1996; Wood, 2000).

Brands tended to be associated with products but there has been a refocus on corporate brands (Mottram, 1998). A strong and favorable corporate brand is seen as an important discriminator in an increasingly competitive environment (Balmer, 1995). According to Ackerman (1998), the corporate brand offers managers a comprehensive discipline for clarifying, humanizing, organizing, and communicating how the company creates value. In industrial markets, the company itself is often the brand; but in consumers markets, the emphasis is usually on the products or a limited group of them (Hague & Jackson, 1994). It would be expensive for industrial companies to brand every item in their wide product range. For many industrial companies, there is scope for only one brand and that is the company name (Hague & Jackson, 1994). According to De Chernatony and McDonald (1998), industrial buyers are primarily concerned with the company's overall brand identity rather than with the specific product they want to buy. For a recent literature review of branding in B2B marketing, see Mudambi (2002).

Studies on industrial branding show mixed results. Saunders and Watt (1979) studied the attempts made by the man-made fibers industry to overcome the problem of loss of identity by branding at the ultimate consumer level. The study concluded that due to the large numbers of brands available, consumers were confused and really did not know what to expect from the different types of fiber. Sinclair and Seward (1988) reported on a survey of established producers of wood and plywood panels who decided to brand their products. This study concluded that the emphasis on price and availability at the retail level tended to suggest that these products were still being produced on a commodity basis. The high dependence on price suggested a low effectiveness of manufacturers' branding strategies.

More recent studies found that branding in the business-to-business sectors were successful. Shipley and Howard (1993) attempted to gain insights into the use of brand names, the nature of brand-name strategy, and the perceived importance among industrial companies. They concluded that industrial companies perceived benefit from using brand names and that large firms valued the benefits of brand names more highly than small firms. Firth (1997) studied the pricing of accounting services in New Zealand. He found that the use of names of the big name accounting firms resulted in a 4% price premium. In a study published in 2001, Michell et al. (2001) found that industrial companies believed branding to be important and provided

competitive benefits and increased brand equity. Hutton (1997) studied professional buyers in the personal computer, copies, fax machines, and computers floppy disks industries. This study concluded that there was brand equity in industrial markets and there was a brand equity "halo effect" in which brand evaluations transferred from one category to another and that buyers were prepared to pay a premium for their favorite brand.

In 1993, Gordon et al. studied the U.S. electrical circuit breakers market. The setting used for this empirical study was the electrical products and components markets. This study provided conclusive evidence that brand equity was alive and well in the business-to-business sector.

2.2. *Intangible attributes*

The literature that deals with the topic of organizational buying has identified and measured nonproduct characteristics and intangible elements of the buying decision in a number of ways (Mudambi et al., 1997).

Levitt (1995) found that the company's overall reputation was generally more influential than the sales presentation, and that technical personnel in high-risk situations relied more heavily on company reputation than on the presentation.

Lehmann and O'Shaughnessy, (1974) examined, across product types, how U.S. and U.K. purchasing agents rated attributes in choosing a supplier. The 17 attributes rated included basic brand factors such as technical specification, tangible examples of brand augmentation such as training, and even more intangible factors such as overall reputation. Their finding that supplier reputation is one of the most highly rated attributes is particularly interesting since reputation lies at the heart of branding strategy (Mudambi et al., 1997).

Abratt (1986) analyzed the industrial buying process for high-technology products to identify the relative importance of the factors influencing supplier selection. Intangible attributes such as supplier's reputation and perceived product reliability were judged more important than a tangible attribute such as price. Shaw et al. (1989) found that a "sizeable portion of the population of buyers report that vendor-related intangible attributes are more important than product performance attributes." They concluded that promotional activities should reflect this concern.

Tangible aspects are quantifiable by measures such as the number of defects, the product life, the lead time, the number of late deliveries, the technical support, the financial services, and the supplier financial stability.

Intangible aspects include factors such as perceived quality, incomplete or conflicting information about the product, ease of ordering, general reliability, willingness of the company to respond in an emergency, service quality, degree of rapport between customers and service providers, understanding between service providers and customers, company reputation, country of origin, pleasantness, and

trustworthiness of company personnel (Mudambi et al., 1997).

2.3. Proposition development

The key research question that needs to be answered: is there a benefit for suppliers of industrial products to engage in branding their products or company? In order to answer this question, five propositions have been developed from the literature review.

P1: *Brand equity exists in business-to-business markets in the form of buyers' willingness to pay a price premium for their preferred brand (Hutton, 1997).*

P2: *Benefits from brand-loyal industrial buyers include their willingness to recommend the brand to peers and to give special consideration to another product with the same brand name (Hutton, 1997).*

P3: *Perceived quality is the primary brand-equity-generating variable (Michell et al., 2001).*

P4: *The primary source of information for building brand awareness are exhibitions and trade shows (Abratt, 1986).*

P5: *Different groups of decision-making unit role players attach different levels of importance to brands and prefer different communication channels (Ghinghold & Wilson, 1998).*

3. Methodology

The method used to gather information for this study was a conjoint analysis experiment supplemented with a self-administered questionnaire. In conjoint analysis, the respondent is asked to make trade-off judgments, for example to decide whether one feature is desired enough to sacrifice another (Aaker, Kumar, & Day, 1995). Cards (containing a profile) were made up according to a fractional-factorial design principle. The analytical objective is to find a set of part worths for the separate factor levels so that when these are appropriately added, one can find the total utility for each combination (Green & Tull, 1978). The combination with the highest total utility should be the one that originally was most preferred, and the combination with the lowest total utility should have been the one least preferred (Aaker et al., 1995). Conjoint analysis establishes the importance of features, the utility of each level of each feature, and the preferred option for each respondent.

3.1. Selection of the field of experiment

The objective of the first part of the study was to explore empirically the existence of brand equity in a specific business-to-business product setting. The setting used for the empirical study was the market for the

medium-voltage indoor circuit-breaker panels. This equipment is used in industrial premises to distribute electricity at medium-voltage levels. It is likely to be found in all industrial sites where electricity demand is substantial, such as in heavy manufacturing plants, chemical plants, mines, and utilities' substations.

This market can be categorized as being in the mature stage of the product life cycle. Because the market is mature, gains in the share by one manufacturer come at the expense of another (Gordon et al., 1993).

This product was chosen for the following reasons:

- It is not purchased by consumers. Its purchase is exclusively limited to industrial buyers.
- The equipment is of a capital nature and is of such value that its purchase will require more consideration than the purchase of a cheap consumable.
- It is widely used by various types of industries.
- It is readily understood and widely used, a key requirement for the survey because of the broad cross section of respondents (Hutton, 1997). It can be unambiguously defined so as to allow comparison among competing brands.
- There are many competing brands in the market. This is important since "price premium measure is defined with respect to a set of competitors" (Aaker, 1996).

3.2. Derivation of variables

Owing to its exploratory nature, the research was conducted in two separate stages: the first stage was qualitative. It aimed at deriving the variables needed for use in the second stage of data collection.

To achieve this, the researchers conducted in-depth interviews with six experts in the field of manufacturing, marketing, and sales of medium-voltage circuit-breaker indoor panels. These interviews were undertaken in order to identify the factors or attributes that these experts considered as important in the procurement process for the equipment concerned.

The important attributes identified were to be used as dimensions for trade-offs in conjoint analysis. The attributes were brands, prices, delivery, technology used, and spare parts lead times. The alternative brands were ABB, Relay Setting, Reyrolle, and Tamco. There were four price levels, R40,000 (US\$4000); R46,500 (US\$4650); R53,500 (US\$5350), and R60,000 (US\$6000). Delivery alternatives were 3, 4, or 5 months. The two types of technologies used were vacuum or SF6. Spare parts' lead times were 1, 2, or 3 weeks. Sixteen cards were generated to collect data that illustrate the multiple-factor approach to conjoint analysis. Each of the 16 cards involves a combination of five attributes identified that is used to describe an offer of a circuit-breaker panel. Note that a full factorial design would have entailed $4 \times 4 \times 3 \times 2 \times 3 = 288$ combinations (Green & Tull, 1978). Thanks to the special factorial design used,

only 16 cards were needed to collect the respondents' preferences.

3.3. Conjoint analysis experiment

Data for conjoint analysis were collected during face-to-face interviews. Respondents were given the 16 cards each describing a complete product configuration. Respondents were asked to rank-order the profiles in order of preference or intention to buy (Aaker et al., 1995). Respondents were left alone since no further guidance was required. Each respondent spent between 7 and 45 minutes completing the experiment. After the experiment, the order of rankings was recorded.

3.4. Methodology for the "halo effect"

One of the issues that this research investigated was whether the goodwill that certain brands enjoy in their current product lines could be extended to new product lines. The treatments considered for this study measured the consideration that respondents were willing to accord to new product lines introduced by the various brands (Proposition 2). The Kruskal–Wallis test was used to test this proposition.

3.5. Population

The population consisted of decision-making unit (DMU) members of industrial companies in South Africa that purchase medium-voltage electrical equipment. Companies that form part of this population belong to a wide range of industries. They were mines, chemical plants, heavy-manufacturing plants, or suppliers of electricity. The only criterion was that they are heavy users of electricity as consumers or as suppliers.

Because the decision making in the buying center is a joint process, the population includes all categories of role players: buyers, users, influencers, deciders, and gatekeepers (Hutt & Speh, 1998).

During the process of in-depth interviews with specialists, a comprehensive database of companies belonging to the population was accessed. The interviews also provided an indication of companies that had purchased medium-voltage circuit-breaker panels during the last 3 years. Three years was chosen as it was assumed that the buying would still be fresh in the respondents' memories.

3.6. Sample

A list of companies that had purchased medium-voltage circuit-breaker panels during the last 3 years was drawn up. Letters of introduction were sent to a convenience sample of companies in South Africa.

The minimum sample size to be chosen took into account the requirement for conjoint analysis (at least

30) and for Kruskal–Wallis (at least five in each DMU role).

3.7. Questionnaire construction

The questionnaire was divided into five parts. The first part was designed to record the results of the conjoint analysis experiment. It is a record of the sequence of profile preferences. The conjoint design was based on five features (brand, price, delivery time, technology, and leadtime for parts). The following levels were used: brand = four levels (ABB, Relay Setting, Reyrolle, and Tamco); price = four levels (R40,000, R46,500, R53,500, and R60,000); delivery time = four levels (3, 4, 5, and 6 months); technology = two levels (Vacuum, SF6); spares leadtime = three levels (1, 2, and 3 weeks).

This design resulted in 16 cards that respondents were asked to rank in order of preference.

The second part consists of four questions designed to determine the respondent's demographic characteristics. These questions allowed the categorization of respondents into specific groups based on their role played in the buying center (or DMU), their educational backgrounds, and their professional experience.

The remaining parts of the questionnaire were made of statements to be evaluated according to a Likert scale graded from 1 to 5, where 5 = *strongly agree* and 1 = *strongly disagree*.

The third part of the questionnaire was designed to investigate the possibility of extension into a new product line where none of the brands was active in the South African market. It requested the respondent to state their consideration of the new product line if it was introduced by each of the brands (companies).

The fourth part investigated the source of brand equity, and the last part of the questionnaire investigated the means of creating brand awareness and the willingness of respondents to recommend their preferred brands.

3.8. Data collection details

Data collection took place during the same face-to-face session as the conjoint analysis. Together with the conjoint

Table 1
Demographic profile of respondents

Position in the DMU	Experience (years)			Total	Education		Total
	3 to 5	6 to 9	>9		Commercial/financial	Technical	
Buyer	1	1	6	8	5	3	8
Gatekeeper	2	2	5	9	5	4	9
User	3	4	5	12		12	12
Technical	2	1	11	14	3	14	14
Decision maker		1	10	11		8	11
Total				54			54

Table 2
Estimated utilities

Feature	Level	Utility
Brand	ABB	+0.528
	Relay Setting	−0.542
	Reyrolle	+0.292
	Tamco	−0.278
Price	R40,000	+0.704
	R46,500	+0.690
	R53,500	−0.458
	R60,000	−0.935
	3 months	+1.116
Delivery time	4 months	−0.194
	5 months	−0.199
	6 months	−0.722
	Vacuum	+0.634
Technology	SF ₆	−0.634
	1 week	+0.586
Spares lead time	2 weeks	−0.182
	3 weeks	−0.404

analysis cards, each respondent was given a copy of the questionnaire and one envelope in which to seal the answers. Respondents were told before hand to seal the envelope as soon as they had finished answering the questions. Data collected were captured on a spreadsheet.

4. Results

Table 1 shows the profile of respondent.

Sixty executives were approached to participate in this study but three could not because of time constraints. There were 57 respondents that participated in the conjoint analysis experiment and the questionnaire. Three respondents were disregarded due to errors they made in recording their conjoint analysis experiment. The 54 remaining respondents were used in the analysis.

All the respondents had more than 3 years of professional experience. Sixty-eight percent had nine or more years experience. All the respondents had a technical or commer-

Table 3

Utility ranges, total utility, and relative importance of attributes

Attributes	Utility	Range of utility	Relative importance (%)
Price	−0.94, +0.704	1.644	24
Delivery period	−0.72, +1.116	1.836	27
Technology	−0.63, +0.634	1.264	19
Brand	−0.54, +0.528	1.068	16
Availability of spare parts	−0.4, +0.586	0.986	14
Total		6.798	100

cial and/or a financial background. People from technical backgrounds were also in the majority among decision makers while the others were mainly buyers and gatekeepers.

4.1. Importance of attributes

MONANOVA was used to calculate the utilities of the various levels of the different features. These are presented in Table 2.

The relative importance of features was calculated from these utilities. These are illustrated in Fig. 1.

Overall, the delivery period was found to be the most desired attribute, followed by price and technology. Brand has a relative importance of 16%, the availability of spare parts is least at 14%.

Table 3 shows the utility ranges, total utility, and relative importance of attributes.

Table 4 shows the relative importance of attributes per role player category.

Despite the asymmetry of input in favor of ‘delivery period,’ all role players with the exception of decision makers rated the ‘price’ as more important than the ‘delivery period.’ But the aversion shown by decision makers for a delivery period, twice as long as the shortest delivery offered, tilted the balance in favor of short delivery periods. Technical specialists were the only respondents to rank the ‘brand name’ as the most important attribute, equally placed with the ‘price.’ They were the only ones to rank the ‘delivery period’ as the least important attribute. Users

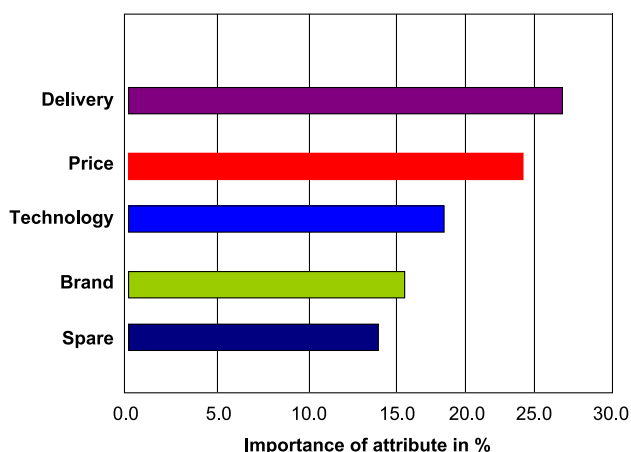


Fig. 1. Relative importance of attributes.

Table 4

Relative importance of attributes per role player category

Attributes	Relative importance for different role players (%)					
	Buyers	Users	Technical specialists	Gate keepers	Decision makers	Total
Price	36	42	24	23	26	24
Delivery period	20	17	14	22	43	27
Technology	16	7	18	28	5	19
Brand	16	28	24	7	19	16
Availability of spare parts	12	6	20	20	7	14

Table 5
Consideration given to other product lines of same brand

	Brand utility	Consideration given to new products ^a	Consideration given to new products ^b
Brand 1 (ABB)	0.528	234	5
Brand 2 (Reyrolle)	0.292	217	4
Brand 3 (Tamco)	– 0.278	174	3
Brand 4 (Relay Setting)	– 0.542	252	3

^a This column represents the sum of scores collected by each brand.

^b This column represents the median value of scores.

awarded the highest relative importance achieved by ‘brand name’ at 28%, just after the price. Overall, ‘technology’ was ranked third at 19%. ‘Technology’ ranked better than brand name and ‘availability of spare parts.’

Respondents were asked to rank the consideration that they would give to new product lines (for example, a medium-voltage electric cable) bearing each one of the four brand names, as considered previously. Table 5 contrasts the utility enjoyed by each of the brands with the willingness to consider new products bearing the same brand name. The willingness is measured both in terms of total scores and the median values. The median values were calculated using the Kruskal–Wallis test.

The brand with the highest utility is given the highest consideration. The consideration given to new product lines of existing brands decreases steadily as the brand’s utility decreases.

4.2. Criteria for selection of preferred brand

The third research proposition deals with the selection criteria for one’s preferred brand. Respondents were asked to rate the importance of nine attributes on a Likert scale from 1 to 5, where 5 = *very important* and 1 = *not important*.

Table 6
Criteria for selection of preferred brand

Criteria	Total score ^a	Median ^b
Quality	260	5
Reliability	259	5
Performance	257	5
After-sales service	232	4
Ease of operation	228	4
Ease of maintenance	227	4
Price	220	4
Supplier’s reputation	216	4
Relationship with Supplier’s personnel	196	4

^a This column represents the sum of scores collected by each brand or company attribute.

^b This column represents the median value of scores awarded to each brand or company attribute.

Table 7
Most effective source of information for brand-awareness creation

	Total score	Median				
		Users	Buyers	Technical specialists	Gatekeepers	Decision makers
Technical consultants	228	4	4.5	4	5	4
Sales representatives	221	4	4	4	4	4
Professional/technical conferences	218	4	4.5	4	4	4
Exhibitions		4	5	4	5	4
Journal/professional magazines	197	3.5	4	4	4	3
Word of mouth	177	2.5	3.5	3	4	4
Direct mail	171	3	4	3	4	3
Mass media	157	3	4	2.5	3	3

Table 6 shows the raw scores for the selection of one’s preferred brand.

Quality, reliability, and performance are listed as the top criteria. Price is third last ahead of supplier’s reputation and relationship with supplier’s personnel.

4.3. Method of brand awareness creation

The method of brand awareness creation is shown in Table 7.

Results in Table 7 rank the use of technical specialists as the most effective means of promoting a new brand. The next in line consists of visits by sales representatives, professional/technical conferences, and exhibitions.

The use of mass media is ranked last, followed by direct mail, word of mouth, and journal/professional magazines. The second column in Table 7 (total score) indicates the sum of scores collected by each medium of communication. Columns 3 to 7 represent the median value of scores collected by each medium of communication.

4.4. Willingness to recommend one’s preferred brand

All the members of the buying center indicated their willingness to recommend their preferred brand. Decision makers and gatekeepers give a rating of 5, buyers 4.5, and users and technical specialists gave a 4 on the five-point scale.

5. Discussion

5.1. Importance of price and delivery

Price and delivery are very important. Together, they explain 51% of the decisions made. Literature that stresses the importance of price and delivery for the industrial

markets abounds (Hague & Jackson, 1994), so much so that a higher score (more than 51%) might have been expected.

Delivery period came out as a more important attribute than price because of the wider range of variation specified for this attribute. It should be kept in mind that “the magnitude of the difference in utilities is strongly influenced by the choices of extreme levels” (Aaker et al., 1995). The extreme attribute levels are 50% for price (R40,000 to R60,000) and 100% for delivery period (3 to 6 months).

The high score achieved by price and delivery must stress that despite the value that is expected from the brand name, price and delivery deserved to be carefully considered. “Responsive pricing strategies must be developed” (Hutt & Speh, 1998).

5.2. Importance of brand name

Brand is said to indicate 16% of the decision. Comparing this with the scores achieved by price and delivery period, it suggests that the difference in preferences across the four brand names is more than half the difference in perception of prices varying by 50% or delivery period doubling. This is a high score. Some industrial projects are won or lost on very tight margins, often below 5%. If a brand name can command a lead of 16% over its competitors, the lead is considerable.

5.3. Price premium for certain brands

Conjoint analysis results indicated clearly that the brand name was not the most important attribute (save for the technical specialists). Delivery time, price, and technology surpassed it. Well-known brand names have positive utility of +0.528 and +0.292, respectively, while lesser known brands have a negative utility of –0.278 and –0.542,

respectively. Price utility ranges from +0.704 to –0.935 for price levels of R40,000 and R60,000, respectively. The higher the price, the lower the utility.

Using linear interpolation, a price of R50,707 would have a utility of zero. The most favored brand (ABB) has a utility of 0.528. Again using linear interpolation and assuming all other things being equal, a utility of –0.528 on price would represent R54,453. Thus, setting the “brand–price” dyad at zero, the most preferred brand could command a price premium of R3746 or 7.4% relative to an “average” price of R50,707. Using similar logic, the least preferred brand (Relay Setting) with a utility of –0.542 would have to discount its price by R3305 to R47,402 or 6.5% relative to the “average” price.

Thus, in this study, the brand leader could afford a price premium of nearly 14% over relatively unknown brands on the market. This is illustrated in Fig. 2.

5.4. Impact of technical specialists on price premiums

A study by Abratt (1986) on industrial buying in high-technology markets demonstrated that “technical personnel dominated the buying process and that the involvement of the buying department was primarily administrative.”

If one were to consider only the opinion of technical specialists, the price premium would have been calculated as follows:

For this group of people, the better known brand has a utility of +1.230, while the least known brand has a utility of –0.928.

To get to neutral utility, the better known brand can increase its price up to R64,020 (utility –1.23). This means that the technical specialists are prepared to grant a price premium of 26% to the leading brand name.

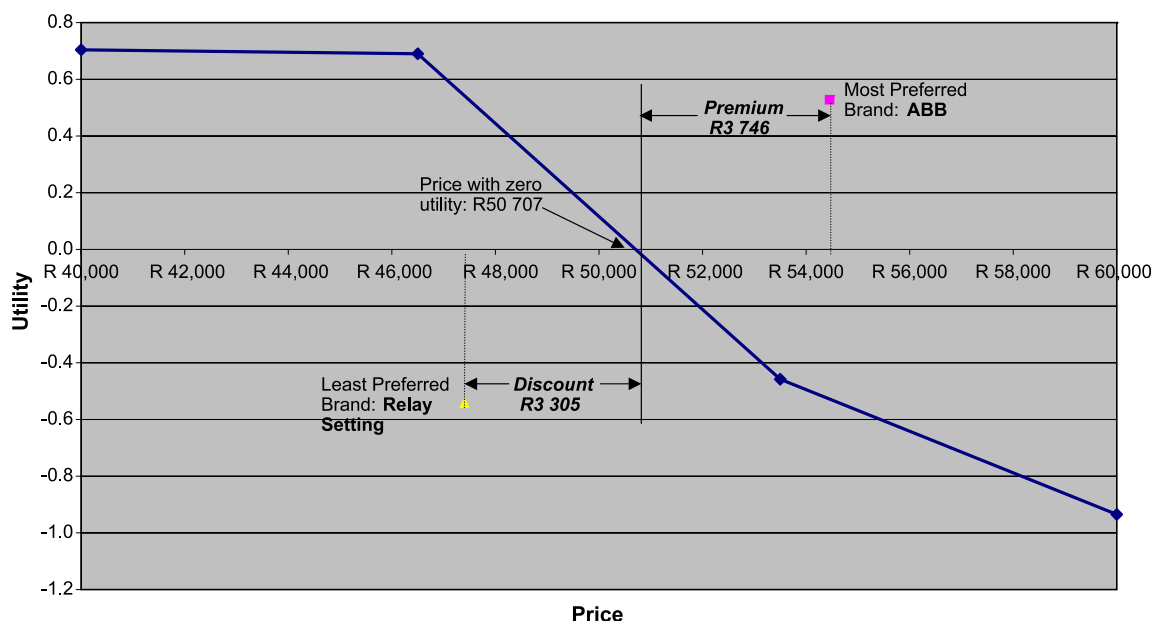


Fig. 2. Brand–price dyad.

5.5. Differences in perception of brand-name importance among DMU role players

Overall, the brand name is said to explain 16% of the choice made. But the perceptions across the DMU role players vary a lot. The users of the product and the technical specialists credit the brand name with higher importance (28% and 24%, respectively).

On the other extreme, the gatekeepers allocate a very low importance to the brand name. This indicates that gatekeepers have difficulty making the distinction between different brand names of specialized technical equipment. This should have been expected because technical brands are outside the area of interest of gatekeepers. The buyers seem to have the same problem but to a lesser degree. The decision makers are somewhere between these extremes.

It is not surprising that users attach so much importance to the brand name. They are the ones who will work with the product in the future. Their previous experience with various brand names will make them more judgmental than others. The same can be said of the technical specialists but to a lesser extent.

5.6. The halo effect: possibility of brand extension into new product lines

The halo effect is the fact of drawing a general impression about an individual on the basis of a single characteristic (Robbins, 1998). In the context of brand, the halo is exploited for brand extensions.

Respondents were asked to say how much consideration they would give to a new product line (medium-voltage electrical cables) in case the four brands decided to enter that market. At the time of the survey, none of the four brands was marketing that line of products.

Responses were given on a scale from 1 to 5. Results indicated that the brand that received the highest score is the same brand with the highest utility and also has the highest median value of consideration. The second ranked brand on sum of consideration is also second on utility and median value. The same applies to the third and fourth brand. The ranking of brands according to consideration received is the exactly the same as the ranking according to brand utility.

Furthermore, the values received by brands were subjected to the Kruskal–Wallis test. The null hypothesis ' H_0 = all medians are equal' states that cables from all four brand names will receive equal consideration. The null hypothesis was rejected at the .05 significance level. Some brands enjoy better prospects of product line extensions than others. Brands with better renowned names have a positive halo effect than lesser known brands.

If a brand has a value, it should be possible to transfer those values to additional products, thus enabling a company to broaden its range. A strong brand can make it easier to launch new products. The original brand will have associ-

ations that will be helpful to the new product (Hague & Jackson, 1994).

5.7. Perceived quality as a brand-generating variable

Quality emerged as the leading criterion for selection of a brand. Other surveys carried out revealed that quality was the most important criteria of selection (Aaker, 1991). Quality ranked also very high (in second position) in the survey conducted by Abratt (1986). Quality, reliability, and performance occupy the same first, second, and third positions, respectively, as they did in the research by Michell et al. (2001).

Consistently, perceived quality is mentioned as the important criterion. That is certainly one of the reasons why it is a brand association that has been elevated to the status of a brand asset (Aaker, 1996).

Aaker (1991) suggests that perceived quality can be defined as the "customer's perception of the overall quality or superiority of a product or service with respect to its intended purpose, relative to alternatives." Perceived quality is, first, a perception by customers. It thus differs from several related concepts, such as

- actual or objective quality;
- product-based quality; and
- manufacturing quality.

Perceived quality cannot necessarily be objectively determined, partly because it is a perception and also because judgments about what is important to customers are involved (Aaker, 1991).

Perceived quality is an intangible, overall feeling about a brand. However, it will usually be based on underlying dimensions, which include characteristics of the products to which the brand is attached such as reliability and performance (Aaker, 1991).

The results of this research ranked reliability and performance second and third, just after the quality. The two attributes are underlying dimensions of quality (Abratt, 1986).

5.8. Price

Abratt (1986) showed that for high-technology products, price was not the most important criterion. The reliance on price as criteria for sales is a characteristic of commodity products (Sinclair & Seward, 1988). The low ranking given to price indicates that industrial equipment is sufficiently differentiated to be discriminated using factors other than price. It has been acknowledged that pricing is a critical element in marketing and competitive strategy. According to Shipley and Jobber (2001), pricing decisions should be based on a comprehensive systematic multistage process that examines and integrates the full range of forces that impacts pricing effectiveness. Although pricing is not the

most important criterion for high-technology products, it should not be ignored. A systematic approach or framework for pricing enhances the quality of price decision making by managers.

5.9. Promotion of a new brand

Respondents were asked to suggest the more effective ways of creating awareness around a new brand. The use of technical consultants was chosen as the preferred way of introducing a new industrial brand. Companies rely on consultants to provide the knowledge that the companies do not always have. According to Mudambi (2002), branding plays a more important role in B2B decision making than has generally been recognized. Understanding how customers perceive the company brand will be key to future management decisions. The specific communications mix needed will depend on the situational factors but will include product catalogues, web sites, advertising, trade shows and exhibitions, and personal selling. To make a new brand attractive in this market, communications will have to highlight product quality and augmented services, as well as the company's reputation. This will reduce the perceived risk and uncertainty faced by buyers.

One lesson to be learned from this finding is that when devising their marketing communications strategy, the marketer should consider all stakeholders who might influence the customer's perceptions. The technical consultants appear to exert a particularly strong influence.

The top three means of promotion for brand awareness are the use of sales representatives, professional and technical conferences, and exhibitions.

6. Conclusions and implications for management

Overall, the leading industrial brand name could command a price premium of 6.8% over the average industrial brand and 14% over a new, unknown brand. The technical specialists were willing to pay a price premium of up to 26%. Technical specialists must be specifically targeted because of their propensity to reward more generously the strong brand and the fact that they are sometimes the ones who make the final brand choice decision.

Proposition 1 is supported: Brand equity exists in the business-to-business markets, in the form of buyers' willingness to pay a price premium for their favorite brand.

The other benefits of brand equity were the willingness to extend the brand's halo to other product lines and recommend the brand to others. It was found that all respondents were willing to recommend their preferred brand. Using the Kruskal–Wallis test, it was shown that the consideration given to new products was biased in favor of well-known brands.

Thus, there is support for Proposition 2: Other benefits from brand-loyal industrial buyers include their willingness

to recommend that brand to peers and give special consideration to another product with the same name.

Proposition 3 investigated the principal brand-equity-generating variable. Respondents were asked to choose among nine variables. Quality was chosen as the first variable and Proposition 3 was supported: Perceived quality is the main brand-equity-generating variable.

Proposition 4 examined the best means to create brand awareness among potential customers. Respondents believed that the most effective way was to get technical consultants to talk to them about the new brand. Overall, visits by sales representatives were chosen from the remaining methods. As far as buyers are concerned, exhibitions remained the preferred channel to generate brand awareness. Other groups preferred various other means of communication. Proposition 4 was supported but was limited to buyers: The primary source of information to industrial buyers for industrial brand awareness is exhibitions/trade shows.

Proposition 5 was concerned about differences across role players in the DMU in relation to perceptions of brand importance and preferred channel of communications. Even though the different role players showed different preferences, the results of the Kruskal–Wallis test indicated that the value of medians across the roles were significantly different. There is no significant difference among the role player groups. Proposition 5 is not supported: Different groups of DMU role players attach different relative importance to brands.

The research has shown that even for a standard and interchangeable product, the DMU members develop a sentimental relationship for which they are prepared to pay a price premium. They are willing to extend that goodwill to other products of the same brand name.

This study shows that in B2B marketing,

- Industrial marketers have something to gain by investing into building a likeable, strong, and positive brand image among all stakeholders. It will allow the company to reap (albeit to a lesser degree) the same benefits that consumer marketers enjoy.
- Quality is the main brand-equity-generating variable. Two lessons derive from this finding. The one is that a quality claim may be effective only if there is substance to the claim. Industrial marketers have to make sure their efforts to build a positive brand image are not torn down by poor quality. The second lesson is that to create a quality product is not enough. Industrial marketers have to translate quality into perceived quality.
- Industrial customers deliberately make it difficult for suppliers to determine who is actually making the buying decision. Therefore, industrial marketers must create a positive image to all stakeholders that come into contact with the company. To achieve this, the supplier company must look beyond marketing communication and develop a total corporate communication program to build up the corporate brand.

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